Logit-lens visualization

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Logit-lens on CODI's 6 latent steps  (GSM8K test, N=50)

Procedure: at each (latent step, layer, sublayer) cell at the LAST-TOKEN
position, apply the model's final LayerNorm + LM head to the hidden
state. Record the top-5 tokens + softmax probabilities.

Sublayers shown (in this order in the residual stream):
  resid_pre  : residual stream coming INTO this block
  attn_out   : the attention block's output (added to residual)
  mlp_out    : the MLP block's output (added to residual)
  resid_post : residual stream coming OUT of this block

Heatmaps color cells by mean top-1 confidence. The text in each cell
is the most-common (modal) top-1 token at that cell across 50 problems.

Interpretation guide:
  - High-confidence cells with the SAME modal token = consistent prediction.
  - The progression from resid_pre → attn_out → mlp_out → resid_post within
    a (step, layer) shows what each sublayer is contributing.
  - Across steps at the LAST layer (L11) reveals how the model's
    intended emission evolves through the latent loop.
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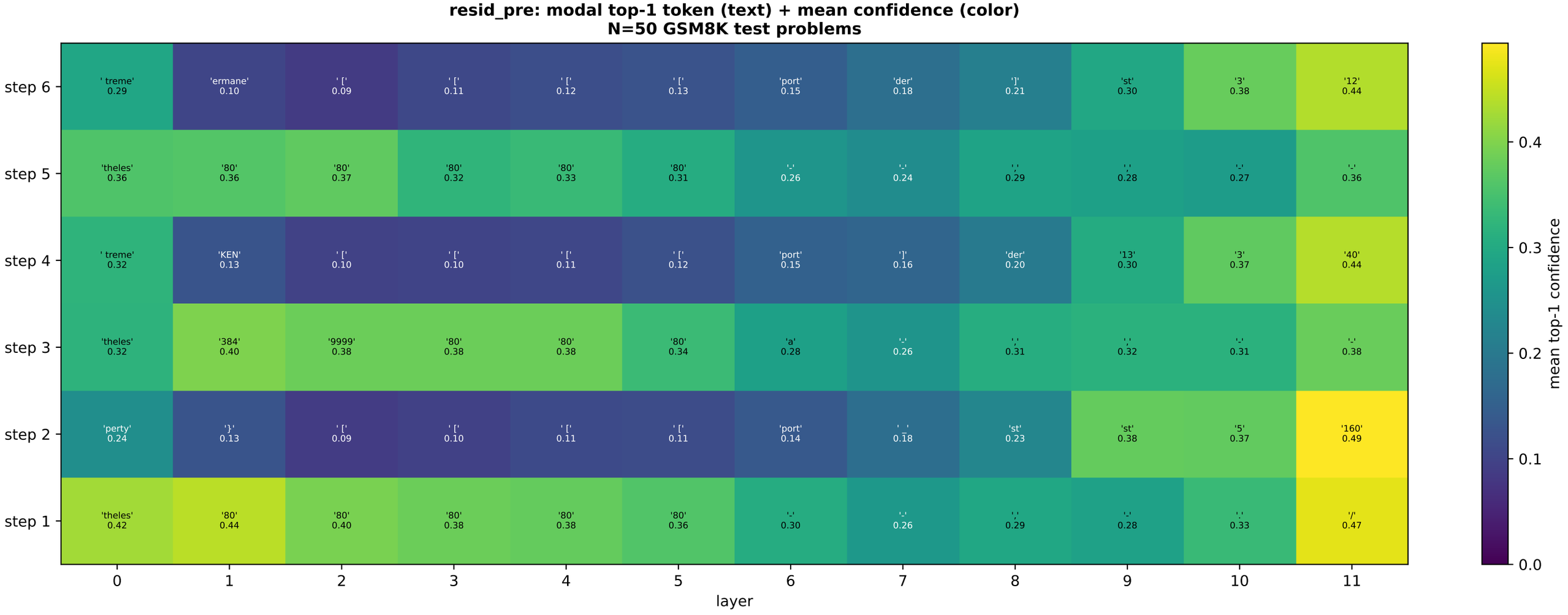
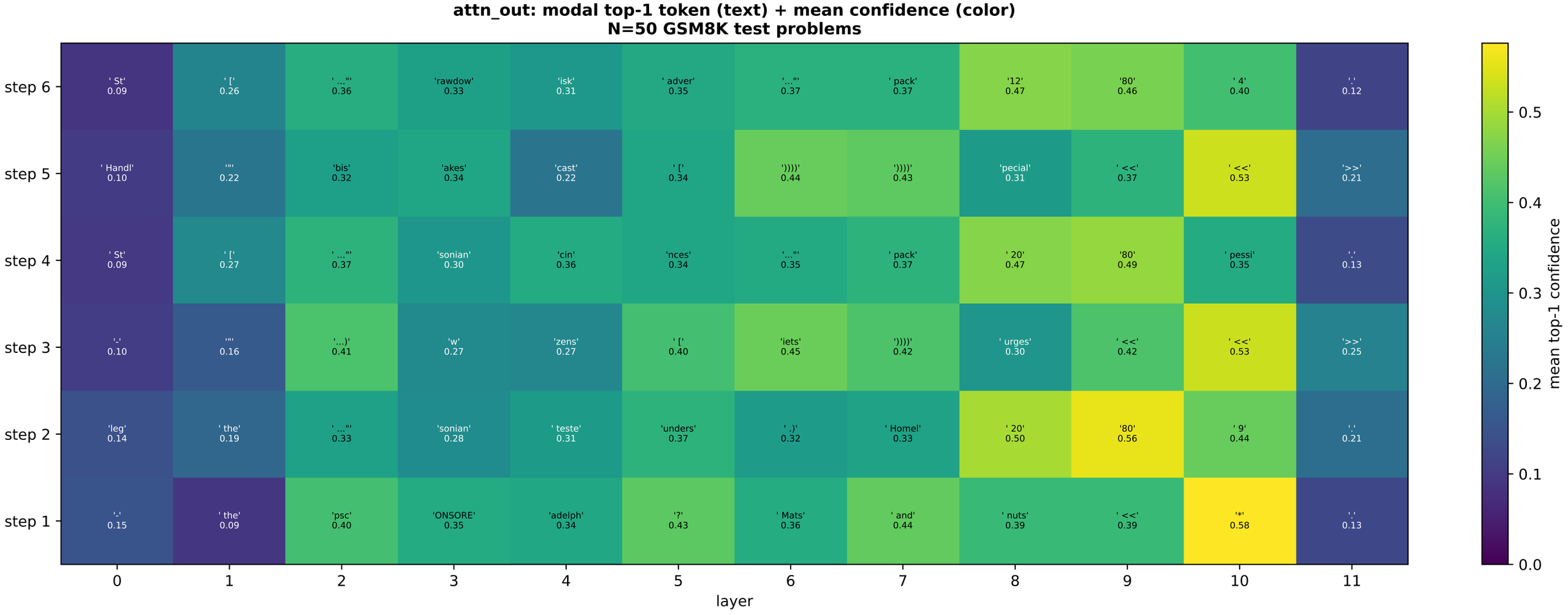
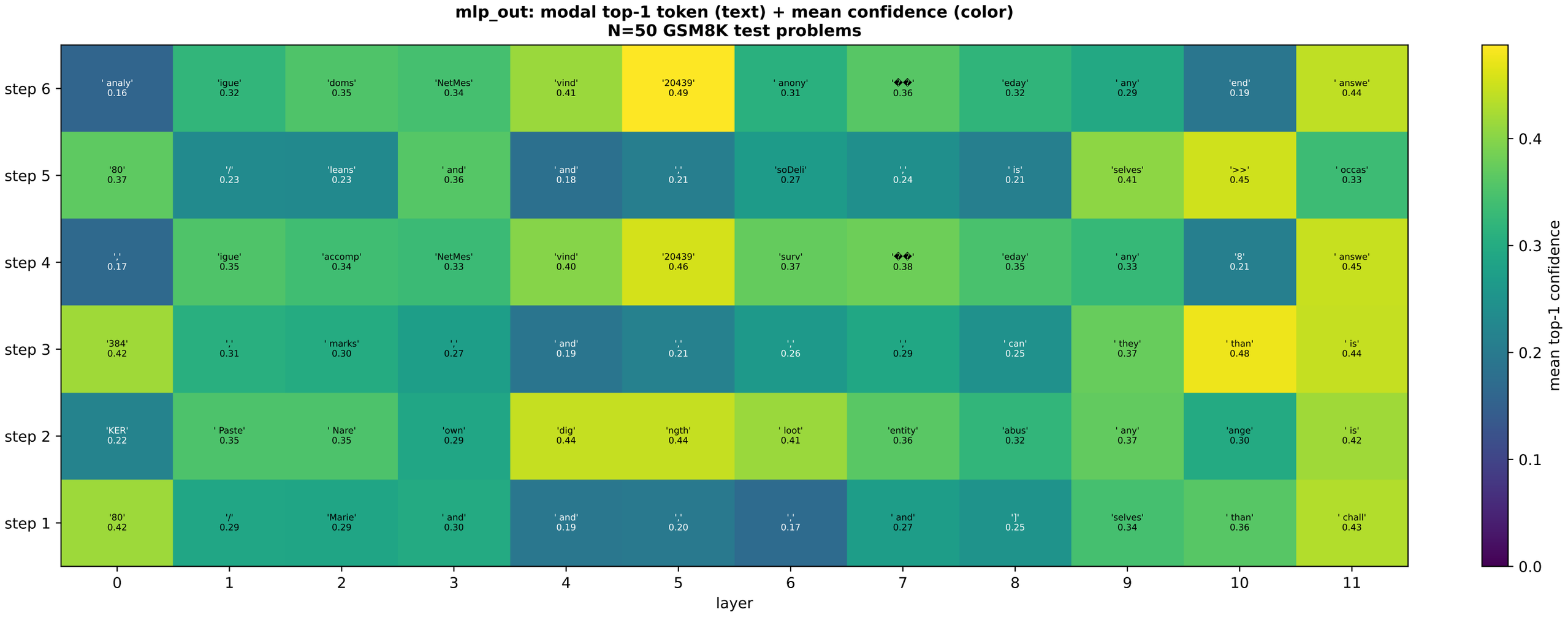
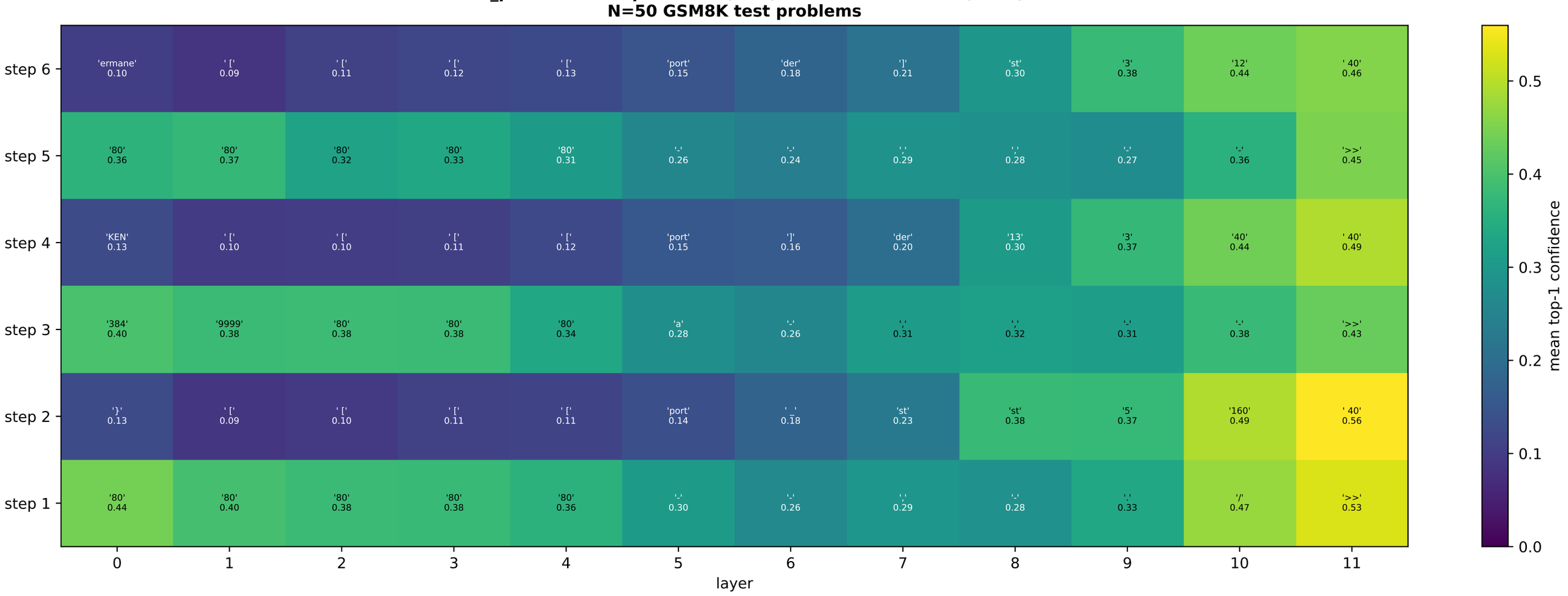
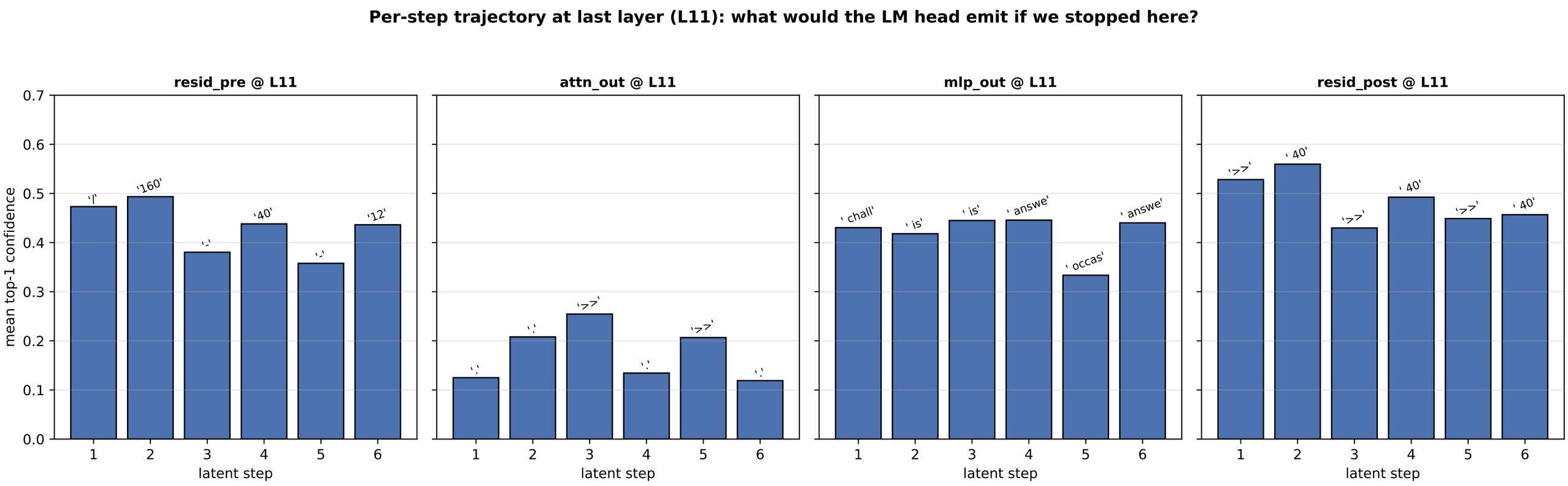

resid_pre: modal top-1 token (text) + mean confidence (color)
N=50 GSM8K test problems
step 6
step 5
step 4
step 3
step 2
step 1
' treme' 0.29
'ermane' 0.10
' [' 0.09
' [' 0.11
' [' 0.12
' [' 0.13
'port' 0.15
'der' 0.18
']' 0.21
'st' 0.30
'3' 0.38
'12' 0.44
'theles' 0.36
'80' 0.36
'80' 0.37
'80' 0.32
'80' 0.33
'80' 0.31
'-' 0.26
'-' 0.24
',' 0.29
',' 0.28
'-' 0.27
'-' 0.36
' treme' 0.32
'KEN' 0.13
' [' 0.10
' [' 0.10
' [' 0.11
' [' 0.12
'port' 0.15
']' 0.16
'der' 0.20
'13' 0.30
'3' 0.37
'40' 0.44
'theles' 0.32
'384' 0.40
'9999' 0.38
'80' 0.38
'80' 0.38
'80' 0.34
'a' 0.28
'-' 0.26
',' 0.31
',' 0.32
'-' 0.31
'-' 0.38
'perty' 0.24
'}' 0.13
' [' 0.09
' [' 0.10
' [' 0.11
' [' 0.11
'port' 0.14
'_' 0.18
'st' 0.23
'st' 0.38
'5' 0.37
'160' 0.49
'theles' 0.42
'80' 0.44
'80' 0.40
'80' 0.38
'80' 0.38
'80' 0.36
'-' 0.30
'-' 0.26
',' 0.29
'-' 0.28
'.' 0.33
'/' 0.47
0
1
2
3
4
5
6
7
8
9
10
11
layer
0.0
0.1
0.2
0.3
0.4
mean top-1 confidence


attn_out: modal top-1 token (text) + mean confidence (color)
N=50 GSM8K test problems
step 6
step 5
step 4
step 3
step 2
step 1
' St' 0.09
' [' 0.26
' ..."' 0.36
'rawdow' 0.33
'isk' 0.31
' adver' 0.35
'..."' 0.37
' pack' 0.37
'12' 0.47
'80' 0.46
' 4' 0.40
'.' 0.12
' Handl' 0.10
'"' 0.22
'bis' 0.32
'akes' 0.34
'cast' 0.22
' [' 0.34
')))' 0.44
')))' 0.43
'pecial' 0.31
' <<' 0.37
' <<' 0.53
'>>' 0.21
' St' 0.09
' [' 0.27
' ..."' 0.37
'sonian' 0.30
'cin' 0.36
'nces' 0.34
'..."' 0.35
' pack' 0.37
' 20' 0.47
'80' 0.49
' pessi' 0.35
'.' 0.13
'-' 0.10
'"' 0.16
'...)' 0.41
'w' 0.27
'zens' 0.27
' [' 0.40
'iets' 0.45
')))' 0.42
' urges' 0.30
' <<' 0.42
' <<' 0.53
'>>' 0.25
'leg' 0.14
' the' 0.19
' ..."' 0.33
'sonian' 0.28
' teste' 0.31
'unders' 0.37
' .)' 0.32
' Homel' 0.33
' 20' 0.50
'80' 0.56
' 9' 0.44
'.' 0.21
'-' 0.15
' the' 0.09
'psc' 0.40
'ONSORE' 0.35
'adelph' 0.34
'?' 0.43
' Mats' 0.36
' and' 0.44
' nuts' 0.39
' <<' 0.39
'*' 0.58
'.' 0.13
0
1
2
3
4
5
6
7
8
9
10
11
layer
0.0
0.1
0.2
0.3
0.4
0.5
mean top-1 confidence


mlp_out: modal top-1 token (text) + mean confidence (color)
N=50 GSM8K test problems
step 6
step 5
step 4
step 3
step 2
step 1
' analy' 0.16
'igue' 0.32
'doms' 0.35
'NetMes' 0.34
'vind' 0.41
'20439' 0.49
' anony' 0.31
'⁇' 0.36
'eday' 0.32
' any' 0.29
'end' 0.19
' answe' 0.44
'80' 0.37
'/' 0.23
'leans' 0.23
' and' 0.36
' and' 0.18
',' 0.21
'soDeli' 0.27
',' 0.24
' is' 0.21
'selves' 0.41
'>>' 0.45
' occas' 0.33
',' 0.17
'igue' 0.35
'accomp' 0.34
'NetMes' 0.33
'vind' 0.40
'20439' 0.46
'surv' 0.37
'⁇' 0.38
'eday' 0.35
' any' 0.33
'8' 0.21
' answe' 0.45
'384' 0.42
',' 0.31
' marks' 0.30
',' 0.27
' and' 0.19
',' 0.21
',' 0.26
',' 0.29
' can' 0.25
' they' 0.37
' than' 0.48
' is' 0.44
'KER' 0.22
' Paste' 0.35
' Nare' 0.35
'own' 0.29
'dig' 0.44
'ngth' 0.44
' loot' 0.41
'entity' 0.36
'abus' 0.32
' any' 0.37
'ange' 0.30
' is' 0.42
'80' 0.42
'/' 0.29
'Marie' 0.29
' and' 0.30
' and' 0.19
',' 0.20
',' 0.17
' and' 0.27
']' 0.25
'selves' 0.34
' than' 0.36
' chall' 0.43
0
1
2
3
4
5
6
7
8
9
10
11
layer
mean top-1 confidence
0.0
0.1
0.2
0.3
0.4


resid_post: modal top-1 token (text) + mean confidence (color)
N=50 GSM8K test problems
step 6
'ermane' 0.10
' [' 0.09
' [' 0.11
' [' 0.12
' [' 0.13
'port' 0.15
'der' 0.18
']' 0.21
'st' 0.30
'3' 0.38
'12' 0.44
' 40' 0.46
step 5
'80' 0.36
'80' 0.37
'80' 0.32
'80' 0.33
'80' 0.31
'-' 0.26
'-' 0.24
',' 0.29
',' 0.28
'-' 0.27
'-' 0.36
'>>' 0.45
step 4
'KEN' 0.13
' [' 0.10
' [' 0.10
' [' 0.11
' [' 0.12
'port' 0.15
']' 0.16
'der' 0.20
'13' 0.30
'3' 0.37
'40' 0.44
' 40' 0.49
step 3
'384' 0.40
'9999' 0.38
'80' 0.38
'80' 0.38
'80' 0.34
'a' 0.28
'-' 0.26
',' 0.31
',' 0.32
'-' 0.31
'-' 0.38
'>>' 0.43
step 2
'}' 0.13
' [' 0.09
' [' 0.10
' [' 0.11
' [' 0.11
'port' 0.14
' _' 0.18
'st' 0.23
'st' 0.38
'5' 0.37
'160' 0.49
' 40' 0.56
step 1
'80' 0.44
'80' 0.40
'80' 0.38
'80' 0.38
'80' 0.36
'-' 0.30
'-' 0.26
',' 0.29
'-' 0.28
'-' 0.33
'/' 0.47
'>>' 0.53
0
1
2
3
4
5
6
7
8
9
10
11
layer
mean top-1 confidence
0.0
0.1
0.2
0.3
0.4
0.5


Per-step trajectory at last layer (L11): what would the LM head emit if we stopped here?
resid_pre @ L11
attn_out @ L11
mlp_out @ L11
resid_post @ L11
mean top-1 confidence
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
latent step
1
2
3
4
5
6
'|'
'160'
'_'
'40'
'_'
'12'
'.'
':'
'>>'
'.'
'>>'
'.'
' chall'
' is'
' is'
' answe'
' occas'
' answe'
'>>'
' 40'
'>>'
' 40'
'>>'
' 40'

Sample 1: top-3 tokens per (step, sublayer) at L11

Problem 1 (gold = 18.0)

Q: Janet’s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fre...

Top-5 tokens at each (step, sublayer) cell at last layer L11:

step	resid_pre	attn_out	mlp_out	resid_post
step1	'-'(0.84),' tha(0.05),'/'(0.02)	' ('(0.05),'*'(0.02),','(0.02)	' pri(0.17),' exp(0.10),'wcss(0.08)	'-'(0.62),'>>'(0.09),' tha(0.04)
step2	' any(0.44),'oo'(0.09),' any(0.08)	'Litt(0.04),'9'(0.03),' 9'(0.03)	' hav(0.11),' -'(0.09),' has(0.06)	' 9'(0.24),'9'(0.06),' any(0.05)
step3	'-'(0.95),'/'(0.00),' tha(0.00)	'>>'(0.06),' <<'(0.05),'<<'(0.04)	' tog(0.91),' ske(0.05),' des(0.02)	'-'(0.87),'>>'(0.02),'NEWS(0.01)
step4	' any(0.23),'oo'(0.14),' any(0.10)	'9'(0.04),'Litt(0.02),' 9'(0.02)	'ogy'(0.05),' 3'(0.03),' -'(0.02)	' 3'(0.08),' 5'(0.08),'9'(0.07)
step5	'-'(0.96),' tha(0.00),'/'(0.00)	'>>'(0.05),'<<'(0.04),' <<'(0.04)	' tog(0.43),'semb(0.16),' des(0.10)	'-'(0.90),'>>'(0.01),' tha(0.00)
step6	' any(0.16),'oo'(0.12),' 5'(0.09)	'9'(0.04),'Litt(0.03),' 9'(0.03)	' -'(0.04),'ogy'(0.03),' 1'(0.03)	' 5'(0.10),' 3'(0.07),'5'(0.07)

Sample 2: top-3 tokens per (step, sublayer) at L11

Problem 2 (gold = 3.0)

Q: A robe takes 2 bolts of blue fiber and half that much white fiber. How many bolts in total does it take?

Top-5 tokens at each (step, sublayer) cell at last layer L11:

step	resid_pre	attn_out	mlp_out	resid_post
step1	'/'(0.36),'st'(0.28),'.'(0.13)	'<|en(0.04),'.'(0.04),'\n'(0.03)	' est(0.30),'<|en(0.22),' inc(0.16)	'>>'(0.66),'/'(0.19),'.'(0.09)
step2	'1'(0.46),' 1'(0.37),'0'(0.15)	'<|en(0.23),' 1'(0.21),'>>'(0.03)	'1'(0.41),'http(0.15),'und'(0.10)	' 1'(0.50),'1'(0.50),' 0'(0.00)
step3	'st'(0.59),')'(0.15),','(0.10)	'>>'(0.44),'ug'(0.02),'<|en(0.02)	' est(0.30),' inc(0.07),' 0'(0.05)	'>>'(0.93),')'(0.03),','(0.01)
step4	'1'(0.51),' 1'(0.36),'0'(0.11)	' 1'(0.39),'<|en(0.12),'>>'(0.04)	'1'(0.51),'http(0.19),' 1'(0.08)	'1'(0.51),' 1'(0.48),' 0'(0.00)
step5	')'(0.70),'st'(0.20),'),'(0.02)	'>>'(0.47),'<|en(0.02),' pat(0.01)	' mat(0.17),' est(0.16),' 0'(0.13)	'>>'(0.93),')'(0.06),'/'(0.00)
step6	'1'(0.57),'0'(0.21),' 1'(0.20)	'<|en(0.18),' 1'(0.16),'>>'(0.04)	'1'(0.52),'und'(0.10),'http(0.09)	'1'(0.67),' 1'(0.32),'0'(0.00)

Sample 3: top-3 tokens per (step, sublayer) at L11

Problem 3 (gold = 70000.0)

Q: Josh decides to try flipping a house. He buys a house for 8O,OOO*andthenputsin*50,000 in repairs. This increased the value of the house by 150%. How much profit did he make?

Top-5 tokens at each (step, sublayer) cell at last layer L11:

step	resid_pre	attn_out	mlp_out	resid_post
step1	' <<'(0.83),'*'(0.02),'\\.'(0.02)	','(0.02),' the(0.02),'.'(0.01)	'𝕏'(0.35),'%.'(0.13),'ubb'(0.13)	' <<'(0.87),'*'(0.02),'.'(0.01)
step2	' per(0.10),'tm'(0.05),' The(0.05)	' ant(0.13),' PAT(0.08),' bod(0.07)	' ans(1.00),' Ans(0.00),' set(0.00)	' rai(0.15),' ans(0.12),'----(0.08)
step3	' 900(0.21),' <<'(0.14),'0000(0.05)	'>>'(0.25),'ull'(0.17),' res(0.03)	'ob'(0.47),'oco'(0.09),'Ub'(0.06)	' <<'(0.36),'>>'(0.07),' 900(0.06)
step4	' Hig(0.27),' Ext(0.08),' Sta(0.03)	' EM'(0.33),' EF'(0.18),' spe(0.05)	' ans(1.00),' Ans(0.00),' set(0.00)	' rai(0.25),' eff(0.08),' Rai(0.07)
step5	'..'(0.10),' <<'(0.06),'666'(0.04)	'ull'(0.08),'>>'(0.07),' res(0.01)	'rett(0.27),'ombo(0.16),'ob'(0.16)	' <<'(0.20),'666'(0.09),'..'(0.07)
step6	' Hig(0.35),' Ext(0.16),' Exp(0.03)	' EM'(0.17),' EF'(0.15),' ant(0.04)	' ans(1.00),' Ans(0.00),' bul(0.00)	' rai(0.25),' eff(0.10),' Sta(0.06)